COMMERCIAL ITEM DESCRIPTION

CURRENT-VOLTAGE STANDARD

The General Services Administration has authorized the use of this Commercial Item Description.

SCOPE. This Commercial Item Description (CID), describes a current-voltage standard for DC voltage ratio accuracy. The current-voltage provides internal switching for calibrating the 100 mV, 1V, 10V, 100V and 1000 volt ranges of a voltage calibrator to a 10 volt reference source without changing the lead connections. The current-voltage standard is part of the F-15 AN/TSM-131A Electrical Standards Set (ESS) and is rack-mounted in a transportable operation case. This CID represents a minimum requirements for the F-15 ESS current-voltage standard in which only those manufacturers who meet or surpass the requirements are qualified per this CID. Equipment covered by this CID may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be type III, class 5, style F and color R in accordance with MIL-T-28800

APPLICABLE DOCUMENTS

SPECIFICATIONS. The following specification forms a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. This CID makes provisions for the bid sample testing in accordance with MIL-T-28800, paragraph 4.2

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for.

(Copies of specification required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

ORDER OF PRECEDENCE. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained

SALIENT CHARACTERISTICS. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

FSC 6625

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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1. SAFETY. The equipment shall be UL (Underwriters Laboratories) listed and approved and shall comply with the type III safety requirements of MIL-T-28800.

2. PERFORMANCE CHARACTERISTICS. The following electrical specifications are minimum requirements

a. RATIO RANGES. 10.1 and 100:1.

b. RATIO ACCURACY. For temperature variation of less than $\pm 1^{\circ}C$ from the self-calibration temperature (between $18^{\circ}C$ and $28^{\circ}C$) for up to 8 hours following self-calibration.

RANGE	INPUT VOLTAGE	OUTPUT UNCERTAINTY	NULL ACCURACY
10:1	100V	0.2 ppm	±0.5 microvolt
100:1	1000V	0.5 ppm	±1.0 microvolt

c. TEMPERATURE COEFFICIENT. Less than or equal to ±1 ppm per $^{\circ}C$ from self calibration temperature over the range of $18^{\circ}C$ to $28^{\circ}C$.

d. INPUT RESISTANCE.

10:1 ratio 380 K ohms ±1%
100:1 ratio Divider is 4 M ohms; Driven Guard is 4 M ohms; total is 2 M ohms ±1%.

e. MAXIMUM INPUT VOLTAGE. 200 volts for the 10:1 ratio and 1100V for the 100:1 ratio.

f. POWER COEFFICIENT. Less than or equal to 0.05 ppm of output with 100V applied for 10:1 ratio and less than or equal to 0.3 ppm of output with 1000V applied for 100:1 ratio.

g. TEMPERATURE. $0^{\circ}C$ to $50^{\circ}C$ operating.

h. RELATIVE HUMIDITY. Less than or equal to 75% to $40^{\circ}C$, less than or equal to 45% to $50^{\circ}C$ non-condensing.

i. SIZE Not to exceed 7 inches h X 8.69 inches W X 24 inches D.

j. WEIGHT. Not to exceed 50 lbs.

3. COMPATIBILITY. This equipment shall be mechanically, electrically and physically compatible with other equipment and accessories of the ESS as the ESS is designed as a measurement entity. The current-voltage standard shall be capable of rack mounting in a standard 19 inch rack. The current-voltage standard shall occupy a vertical rack height not to exceed 7 inches.

4. MANUALS. The manufacturer shall furnish one copy of instruction manuals for each instrument in this equipment group. It shall contain at least the following: Operating Instructions, Theory of Operation, Circuit Description, Schematics, Trouble Shooting Trees, Pictorial Layouts, Parts Locator Diagrams,

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List of Replaceable Parts, Part Numbers, and Manufacturers. In addition, all internal major components, assemblies, subassemblies, and plug-ins, shall be permanently labeled in accordance with manuals, diagrams and schematics furnished.

5. CALIBRATION INTERVAL. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

CERTIFICATION. The contractor shall certify that the product offered meets the salient characteristics of this description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices, and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

PACKAGING PACKING AND MARKING. The items shall be packaged in accordance with normal commercial practices and packed to assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling and storage. The shipping container shall be in compliance with national motor freight classification and uniform freight classification. Shipping containers shall be marked as specified in the contract or order.

KNOWN ACCEPTABLE PRODUCTS. The following product(s) are known to meet the requirements of this CID.

MANUFACTURER	MODEL	NSN
John Fluke Mfg Co., Inc.	752A	6625-01-175-1192

Custodian:
Air Force - 99
Navy -EC
DLA - ES

Preparing activity:
Air Force - 82

Reviewing activity:
Army - CR

(Project 6625-F851)

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